Automatic or Controlled? Repetition Priming Reveals Divergent Processing in Base LLMs, Instruct LLMs, and Humans

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Abstract

Words recur constantly in natural language use, yet it remains unclear whether language models reactivate prior representations or re-evaluate repeated words afresh, and whether post-training changes this default behavior. In this paper, we apply repetition priming (Shiffrin & Schneider, 1977) to 15 models across 5 families (1.5B–14B parameters) in two tasks (semantic categorization and cloze completion), with matched human experiments using identical stimuli. Our findings reveal that: (1) Base models exhibit *automatic processing*, showing immediate facilitation that is stable across lags, partially survives context removal, and directly correlates with attention to prior occurrences; (2) Instruct models exhibit *controlled processing*, showing facilitation that decays with lag, collapses without expected context, and at larger scales reverses to interference; (3) The dissociation widens monotonically with scale within the Qwen 2.5 family (1.5B–14B), suggesting that post-training progressively suppresses automatic facilitation; (4) Humans show a hybrid profile that is lag-sensitive like instruct models but uniformly facilitative like base models, never reversing to interference. Our study demonstrates that the post-training pipeline is associated with a qualitative shift in how models process repeated information, provides mechanistic evidence for where the two modes diverge, and reveals that human cognition occupies a middle ground that neither model type captures.

1 Introduction

Words recur constantly in natural language use, including in multi-turn dialogues, overlapping retrieval passages, and repeated few-shot exemplars (Brown et al., 2020; Ouyang et al., 2022). When a language model encounters a word it has already processed in the current context, does it simply reactivate the prior representation, or does it re-evaluate the word afresh? We present evidence that the answer is associated with post-training, including supervised instruction tuning and, for some models, preference optimization. Base models default to fast reactivation: repeating a word facilitates processing regardless of intervening context. Instruct models engage in context-dependent re-evaluation: facilitation decays with lag and, at scale, reverses to interference. The difference is not merely quantitative but reflects two qualitatively distinct processing modes.

To characterize these mechanisms, we draw on the *automatic vs. controlled processing* framework from cognitive psychology (Shiffrin & Schneider, 1977). Automatic processing is fast, stimulus-driven, and robust to interference; controlled processing is slower, context-dependent, and sensitive to disruption. Repetition priming, a well-established paradigm from human psycholinguistics (Scarborough et al., 1977; Forster & Davis, 1984; Tenpenny, 1995), offers a controlled way to probe these mechanisms: by varying the lag between repeated exposures, we can test whether facilitation is durable and stimulus-driven or transient and sensitive to contextual interference. Recent work has begun applying dual-process frameworks to LLM behavior, framing the shift from fast heuristic responses to deliberate reasoning as a System 1 to System 2 transition (Brady et al., 2025; Li et al., 2025; Ziabari et al.,

2025), but primarily through prompting manipulations. We take a different approach: using repetition priming to show that the post-training pipeline is associated with a qualitative shift in the default processing mode.

We test 15 models across 5 families (1.5B–14B parameters) in two complementary experiments: *semantic categorization* (Experiment 1), which probes how repeated exposure affects forced-choice classification, and *cloze completion* (Experiment 2; Taylor, 1953; Peelle et al., 2020), which probes how repeated exposure aids open-ended word retrieval from weakly constraining contexts. Matched base/instruct variants provide a controlled within-family contrast, holding architecture and pretraining family as constant as public checkpoints permit. For each experiment, **we conduct matched human experiments with identical stimuli and design**.

Across both tasks, we find a consistent dissociation. Base models show strong, immediate facilitation that is stable across lags and robust to intervening material. Instruct models show weaker facilitation that decays with lag and, at larger scales, reverses to interference—the same word that helped the model earlier now hurts it. This is not a subtle quantitative difference: it is a qualitative split between two processing modes, suggesting that instruction tuning does not merely reshape *what* models output but fundamentally changes *how* they process repeated information.

Three mechanistic probes support this interpretation. Removing prior response context produces residual facilitation in base models but interference in instruct models; remapping classification labels shows partial transfer in base models but none in instruct models; and ablating attention heads associated with prior-occurrence retrieval sharply reduces priming only in base models. Together, these results indicate that base-model priming is at least partially word-level and stimulus-driven, while instruct-model priming depends more on downstream contextual evaluation. Within the controlled Qwen 2.5 series (1.5B–14B), the dissociation widens monotonically with scale. Matched human experiments reveal a third profile: lag-sensitive like instruct models but uniformly facilitative like base models, never reversing to interference.

2 Related Work

2.1 In-Context Learning and Repetition

In-context learning (ICL) enables models to adapt behavior based on prompt examples without parameter updates (Brown et al., 2020), with mechanistic studies identifying attention-based retrieval processes such as induction heads (Olsson et al., 2022) and characterizing ICL as implicit optimization (Dai et al., 2023; Akyürek et al., 2023). ICL research focuses on *task-level* adaptation from structured examples; we study *item-level* repetition across intervening material. Lag manipulation distinguishes durable item-specific effects from short-range recency phenomena (Min et al., 2022; Zhao et al., 2021; Liu et al., 2022). Relatedly, Vaidya et al. (2023) find that humans and LMs diverge when predicting repeating text, linking LM repetition to attention heads and recency. Our controlled design isolates the processing consequence of a repeated lexical item and compares matched base/instruct checkpoints.

2.2 Priming in Language Models

Semantic priming studies report partial correspondence between human and model facilitation for related words (Ettinger et al., 2016; Misra et al., 2020), while *structural priming* studies show reuse of recent syntactic constructions (Sinclair et al., 2022). Repetition priming instead concerns the *same* word. Mahaut & Franzon (2025) show that natural and ICL-induced repetition in generation rely on distinct mechanisms, differing in confidence, attention allocation, and sensitivity to perturbation. Their task concerns generated repetition; ours concerns controlled repeated-item processing. Both results show that superficially similar repetition effects can arise from distinct mechanisms. We use the diagnostic toolkit of human repetition-priming research and the automatic vs. controlled framework (Shiffrin & Schneider, 1977) to characterize how post-training shifts the default processing mode.

Model Family	Variants	Params
LLaMA 3	base, instruct	8B
Mistral	base, instruct	7B
Gemma 2	base, instruct	2B
Phi-4	instruct	14B
Qwen 2.5	base, instruct	1.5B
	base, instruct	3B
	base, instruct	7B
	base, instruct	14B

Table 1: Models tested (15 models across 5 families). The Qwen 2.5 family spans four parameter scales, enabling analysis of how model size interacts with instruction tuning to affect information processing.

2.3 Effects of Instruction Tuning

Instruction tuning is known to reshape model behavior beyond surface-level compliance (Ouyang et al., 2022). Wu et al. (2024) show that instruction tuning adjusts self-attention heads to capture instruction-relevant relationships and rotates feed-forward representations toward user-oriented tasks, suggesting deep representational changes rather than superficial formatting shifts. Comparisons of base and instruct variants reveal systematic behavioral differences: base models outperform instruct models in retrieval-augmented generation settings (Cuconasu et al., 2024), instruction-tuned models show improved knowledge acquisition (Jiang et al., 2024), while instruct models show enhanced but also more brittle task-following behavior. These findings motivate our question: does instruction tuning also change how models process repeated information at the item level? We provide evidence that it does, shifting the default from automatic to controlled processing. Extended related work on dual-process frameworks, LLM–human cognitive comparisons, and repetition suppression in neuroscience appears in Appendix A.

3 Experiment 1: Semantic Categorization

3.1 Models

We tested 15 models spanning 5 families and 5 parameter scales (Table 1). At the 7–8B scale, we tested LLaMA 3 (8B), Mistral (7B), and Qwen 2.5 (7B), each with base and instruct variants. We also tested Gemma 2 (2B, base and instruct) and Phi-4 (14B, instruct only). To investigate scaling, we tested Qwen 2.5 at four scales: 1.5B, 3B, 7B, and 14B (base and instruct at each).

All models were run in bfloat16 precision with eager attention for attention matrix extraction.

3.2 Stimuli and Design

We selected 182 concrete nouns from the McRae et al. (2005) feature norms, balanced between living entities (91) and nonliving objects (91). Forty-five non-target concrete nouns served as fillers, balanced across category (22 living, 23 nonliving); fillers did not repeat within a target's intervening window. A control for the proportion of same-category fillers did not predict target priming ($\beta = 0.02$, $SE = 0.05$, $t = 0.41$, $p = .68$) or alter the exposure and lag effects.

Each target word appeared 4 times (exposures E1–E4), with a fixed number of filler trials between consecutive exposures determined by the lag condition (1, 3, 7, or 15 intervening trials). Each lag condition was run as a separate session with context reset before each target word's first exposure, ensuring that priming effects reflect only within-word repetition. This within-item design allows us to measure the change in processing from E1 (baseline) to subsequent exposures.

3.3 Procedure and Measures

On each trial, models received a zero-shot classification prompt and were asked to categorize a word as "living" or "nonliving." Following Petroni et al. (2019), we measured the log-probability margin:

$$\text{Margin} = \log P(\text{correct}) - \log P(\text{incorrect}) \tag{1}$$

The priming effect, our key measure of how repeated exposure changes information processing, was computed as:

$$\text{Priming}_{E_i} = \text{Margin}_{E_i} - \text{Margin}_{E_1} \tag{2}$$

Positive values indicate facilitation; negative values indicate interference.

3.4 Experiment 1a: LLM Results

Priming magnitude. All 7 base models showed strong, uniformly positive priming by the final exposure (E4−E1 range: +5.04 to +6.89, all $p < .001$; Table 7; Figure 1). Instruct models were weaker and more variable: most showed moderate facilitation, but two showed interference (Mistral-Instruct and Qwen2.5-14B-Instruct; see Table 7).

Lag profiles. To estimate the joint effects of exposure number and lag on priming magnitude, we fit a linear mixed-effects model separately for each model.[1]

$$\text{M}_{ij} = \beta_0 + \beta_1 \text{Exp}_i + \beta_2 \text{Lag}_j + (1 + \text{Exp}_i \mid \text{Word}) + \varepsilon_{ij} \tag{3}$$

where M_{ij} is the log-probability margin, Exp_i is ordinal exposure, Lag_j is the number of intervening items, and $(1 + \text{Exp}_i \mid \text{Word})$ allows baseline difficulty and repetition sensitivity to vary by item. All 7 base models showed positive exposure effects (all $p < .001$) and no lag effects. Six of eight instruct models showed negative lag effects at uncorrected thresholds; five remained significant after FDR correction. Facilitation therefore decayed with lag and, in Qwen2.5-14B-Instruct, reversed to interference (Table 8).

Scaling trend. The Qwen 2.5 family (1.5B, 3B, 7B, 14B) provides the only continuous matched size series in our model set (Figure 2). Within this family, base priming was stable across scale, while instruct priming declined monotonically and reversed to interference at 14B. We do not infer from this within-Qwen pattern that size alone explains cross-family variation.

3.5 Experiment 1b: Human Experiment

To anchor the LLM findings in human cognition, we conducted a matched semantic categorization experiment with 40 native English speakers (22 female, 18 male; mean age = 23.4, $SD = 3.1$). Humans classified a subset of 10 target words (5 living, 5 nonliving) using the same lag conditions and per-word context isolation design, with response time (RT) and accuracy as dependent measures.

Human priming: facilitative and lag-sensitive. Humans showed robust facilitation in both RT and accuracy. Mean RT decreased from 648 ms at E1 to 539 ms at E4 ($t(39) = 11.82$, $p < .001$, $d = 1.87$), and accuracy improved from 92.3% at E1 to 97.1% at E4 ($t(39) = 3.28$, $p = .002$, $d = 0.52$). Critically, human priming showed lag-dependent decay in RT ($\beta = -1.72$, $t = -5.84$, $p < .001$), well-described by a power-law function consistent with McKone (1995). Accuracy priming was less lag-sensitive ($\beta = -0.08$, $t = -1.42$, $p = .16$), likely reflecting a ceiling effect. Humans also showed a significant low-frequency advantage in RT (131 ms vs. 87 ms, $p = .003$).

[1] Per-model statistics are reported for transparency; our claims rest on the consistency of the qualitative pattern across model families, not on individual p-values.

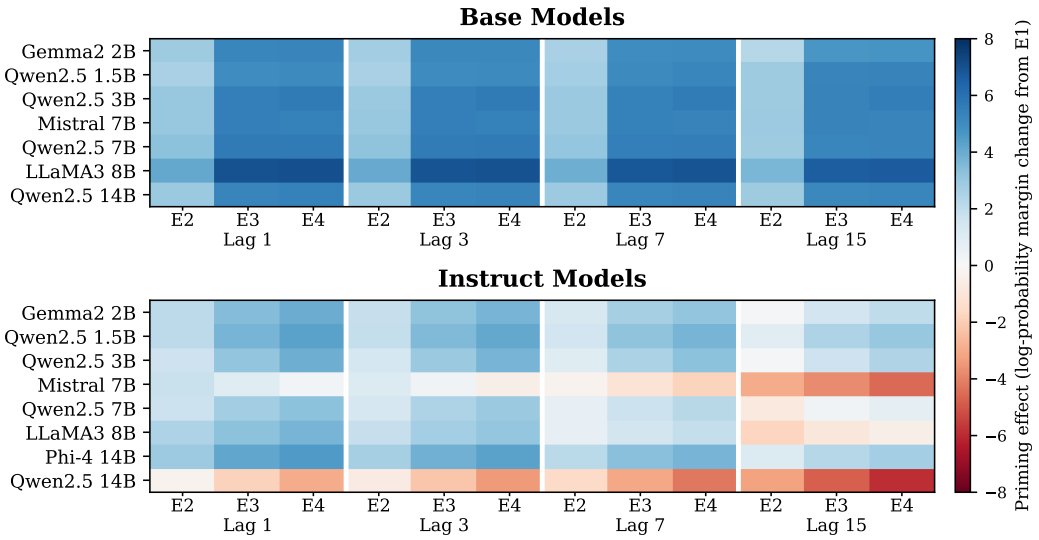


Figure 1: Experiment 1 priming profiles across lag conditions and exposures. Base and instruct models are shown in separate enlarged panels; blue indicates facilitation and red indicates interference. Numerical exposure means and fitted lag coefficients are reported in Appendix Tables 7 and 8.

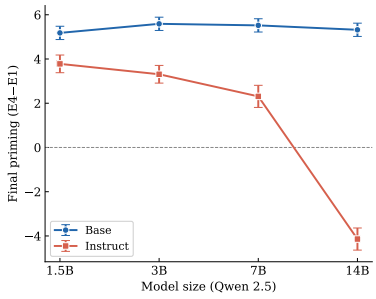


Figure 2: Experiment 1 scaling trends (Qwen 2.5). Final priming (E4−E1) is stable for base but declines for instruct models. Error bars: SEM.

Comparison with LLM patterns. Human lag sensitivity resembles the instruct model pattern, not the base model's lag invariance. However, human priming was uniformly facilitative (never producing interference), unlike the interference observed in some instruct models. The low-frequency advantage in humans also contrasts with the null frequency effect in LLMs. We return to these convergences and divergences in the Discussion (Section 6).

4 Experiment 2: Cloze Completion

Experiment 1 established the processing mode dissociation in forced-choice classification. Experiment 2 tests whether the same dissociation extends to open-ended word retrieval, where models must recover a specific lexical item from weakly constraining sentence contexts, providing a purer test of how repeated exposure changes information retrieval.

4.1 Stimuli

We used standardized completion norms from Peelle et al. (2020) (3085 English sentence contexts normed by 200+ participants). We selected 21 target words, each paired with 4 low-cloze sentence frames (cloze probability: 0.05–0.15). At these cloze levels, the sentence context is weakly constraining: many words are plausible completions, so no single word

dominates. For example, "James had to see his doctor about his _____" (target: *rash*, cloze = 0.09) admits *back*, *leg*, *cough*, etc.; "Greg looked down and realized he was missing his _____" (target: *ring*, cloze = 0.15) admits *wallet*, *keys*, *phone*, etc. Using exclusively low-cloze sentences ensures that the sentence alone weakly constrains the target, so improved performance can only arise from prior in-context exposure, making this a direct test of information retrieval mechanisms.

Target words satisfied three constraints: (1) $\geq$4 low-cloze sentences available, (2) single-word completions, and (3) zero contamination, meaning the target word does not appear in any other sentence. The 21 words comprised 14 nouns (*church*, *fear*, *friend*, *horse*, *mud*, *names*, *nose*, *rash*, *rat*, *ring*, *salt*, *song*, *speed*, *spider*), 6 verbs (*fight*, *laugh*, *lure*, *sail*, *steal*, *wash*), and 1 adjective (*sour*); full list in Appendix C. We selected 50 filler sentence–word pairs verified to contain no target word; the same fillers were reused across lag conditions.

4.2 Design and Procedure

Each target word appeared 4 times (E0–E3) across different sentence frames, with 1, 3, or 7 filler trials between exposures. Context was reset before each word's first exposure (E0), ensuring that priming effects reflect only within-word repetition. A baseline session presented each sentence in isolation.

4.3 Analysis

The dependent measure parallels Experiment 1: we track how the log-probability assigned to the target word changes across repeated exposures. Because cloze completion is open-ended (no fixed incorrect alternative), we use the raw log-probability of the target rather than a margin between correct and incorrect options.[2] Priming is defined as:

$$\text{Priming}_{E_i} = \log P(\text{target})_{E_i} - \log P(\text{target})_{E_0} \tag{4}$$

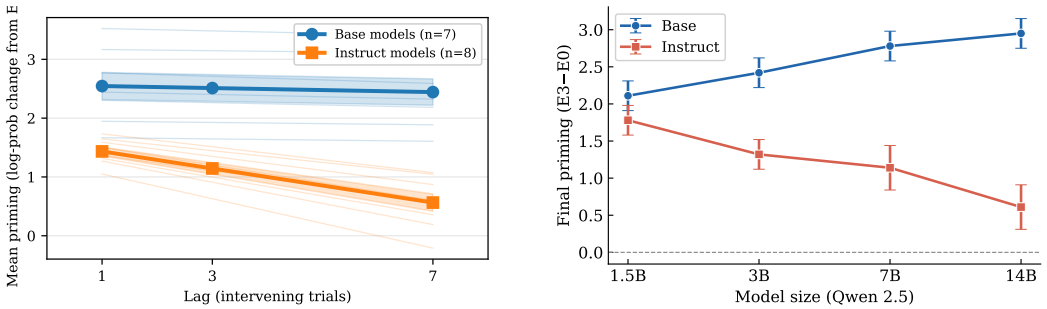
where E_0 is the first in-session exposure. We use E0 rather than the isolated baseline session as the reference because E0 reflects the model's response to the target word within the same context window and session format as subsequent exposures, providing a cleaner within-session comparison. The isolated baseline session serves as a verification that E0 performance is comparable to truly naive performance (which it is: mean difference < 0.05 log-probability units across models). We report priming at each subsequent exposure (E1$-$E0, E2$-$E0, E3$-$E0) and fit the same mixed-effects model structure as Experiment 1 (Equation 3), replacing margin with log-probability as the dependent variable and using lag $\in \{1, 3, 7\}$.

4.4 Experiment 2a: LLM Results

Priming magnitude. All 7 base models showed strong positive priming (E3$-$E0 range: $+2.11$ to $+4.12$, all $p < .001$; Table 9; Figure 3a). Instruct models showed consistently weaker effects (E3$-$E0 range: $+0.61$ to $+1.85$), with the base/instruct gap present across all families.

Lag profiles replicate the Experiment 1 dissociation. All base models showed significant exposure effects with no lag sensitivity, while all eight instruct models showed negative lag effects at uncorrected thresholds; six remained significant after FDR correction (Table 10). Omnibus mixed-effects models confirmed ModelType$\times$Lag interactions in Experiment 1 ($\beta = -0.26$, $SE = 0.04$, $t = -6.50$, $p < .001$) and Experiment 2 ($\beta = -0.13$, $SE = 0.03$, $t = -4.52$, $p < .001$). Main between-type effect sizes ranged from $d = 1.82$ to 2.93 (Appendix F).

[2] Raw log-probabilities may partly reflect sentence-level entropy differences across frames. Our within-item design (each word serves as its own baseline at E0) controls for this, since priming is computed as a within-word change. We verified that results are qualitatively unchanged when using rank-normalized probabilities.



(a) Mean priming by lag. Base models (blue) maintain stable priming; instruct models (orange) decline with lag.

(b) Scaling trends (Qwen 2.5). Final priming (E3−E0) improves with scale for base models but declines for instruct models.

Figure 3: Experiment 2 results. (a) Thin lines: individual models; thick lines: group means. Error bars/shaded regions: SEM. (b) Error bars: SEM.

Prompt robustness. Re-running Experiment 1 with three semantically equivalent instruction/answer-cue templates preserved the LLaMA3 base/instruct split: base lag slopes ranged from -0.04 to $+0.01$ (all n.s.), versus -0.32 to -0.27 for instruct (all $p < .001$); the Prompt$\times$ModelType$\times$Lag interaction was not significant ($F = 1.12$, $p = .33$). Analysis details appear in Appendix D.

Scaling trend. Within the Qwen 2.5 family, base model priming improved with scale while instruct model priming declined (Figure 3b), replicating the Experiment 1 scaling pattern.

4.5 Experiment 2b: Human Experiment

Another 38 native English speakers (20 female, 18 male; mean age = 22.8, $SD = 2.9$) completed a matched cloze experiment. On each trial, participants saw a sentence frame with a blank and typed a single-word completion. In the *study phase*, participants read a complete sentence containing the target word (e.g., "The man was professionally trained in how to **fight**."). In the *test phase*, after 1, 3, or 7 filler trials, participants completed a different sentence frame for the same target word (e.g., "Afraid of getting injured, Zach backed out of the _____."). Dependent measures were *target production rate* (whether the participant produced the target word) and *response time* (RT). A baseline condition presented test sentences without any prior study exposure.

Human priming: gradual and lag-sensitive. Prior exposure significantly increased target production rate: from 11.2% at baseline to 28.4% after one study exposure ($\chi^2(1) = 18.7$, $p < .001$), rising to 41.7% after three exposures. RT for correctly produced targets also decreased with exposure (from 4.82 s at E1 to 3.41 s at E3; $t(37) = 4.93$, $p < .001$, $d = 0.78$). Both measures showed significant lag-dependent decay (production rate: $\beta = -2.1\%$ per lag unit, $z = -3.42$, $p < .001$; RT: $\beta = +0.18$ s per lag unit, $t = 2.87$, $p < .01$). This pattern resembles the instruct model trajectory: gradual, lag-sensitive improvement rather than the base model's near-instantaneous retrieval.

Comparison with LLM patterns. As in Experiment 1b, human priming was lag-sensitive (resembling instruct models) but uniformly facilitative. Even at the longest lag, human target production remained above baseline (19.8% vs. 11.2%, $p < .01$). RT and model log-probability are not directly commensurate; we compare only the profile—facilitation versus interference, lag sensitivity, and reversal (Appendix Table 5).

5 Mechanistic Evidence

The preceding experiments establish *that* base and instruct models differ; the following analyses probe *how*, by selectively disrupting contextual information and internal retrieval pathways.

5.1 Evidence 1: Missing Context Reveals Opposite Processing Defaults

What happens when models encounter a repeated word but the prior response is missing? We tested this in Experiment 1 by replacing classification responses ("living"/"nonliving") in the context history with placeholder tokens ("...") while maintaining word repetition. (This ablation is not applicable to Experiment 2, where removing the target word completion would eliminate the word's presence in context entirely, precluding any repetition effect.)

Results. Base models showed reduced but still positive priming (mean: $+0.70$; range: $+0.41$ to $+1.38$), approximately 15% of the full-context effect. The repeated word still triggered partial activation even without response history, consistent with stimulus-driven processing. Seven of eight instruct models shifted to negative priming (range: -1.26 to -4.84; overall instruct mean: -2.36). The exception, Qwen2.5-14B-Instruct, showed positive priming at lag 1 that reversed at longer lags (mean -0.61; see Appendix Table 11). Without the expected response context, controlled processing generates interference: the model detects the repeated word but interprets the incomplete context as evidence against its prior assessment.

This asymmetric response to missing context distinguishes the two processing modes. The same manipulation produces weakened facilitation from the automatic system (the stimulus alone still partially activates its representation) versus active interference from the controlled system (expected context missing, generating conflict).

Converging evidence from label remapping. We reversed the classification labels after Phase 1 training (living→A became living→B) and tested whether priming survived. Among base models, 3 of 7 showed significant facilitation despite reversed labels, with the remaining four showing non-significant positive trends. Among instruct models, none showed significant transfer. This confirms that base model priming is at least partially word-level, while instruct model priming is tied to the specific stimulus-response context (details in Appendix H).

5.2 Evidence 2: Attention Patterns Suggest Shared Retrieval with Divergent Downstream Processing

The preceding evidence establishes *behavioral* differences. Does the underlying retrieval mechanism also differ, or do both model types retrieve prior occurrences but process them differently? We examine this question through attention weight analysis across both experiments. We measured attention to positions where the target word had previously appeared. The *attention ratio*, defined as attention to prior occurrences divided by the uniform-distribution expectation, quantifies whether models preferentially attend to prior context (Figure 4, panels a–b).

Both model types attend to prior occurrences. Across both experiments, base models showed consistently elevated attention ratios, while instruct models were more variable (Figure 4, panels a–b). Ratios were lower in Experiment 2, consistent with the longer cloze contexts distributing attention more broadly. Most models attended to prior occurrences above baseline, suggesting that attention-based retrieval is a *shared* mechanism.

The divergence appears downstream. The attention–behavior relationship differed between model types (Figure 4, panels c–d). Base models showed significant positive correlations between attention to prior occurrences and priming magnitude in both experiments, indicating that retrieval directly drives facilitation. Instruct models showed uniformly weak

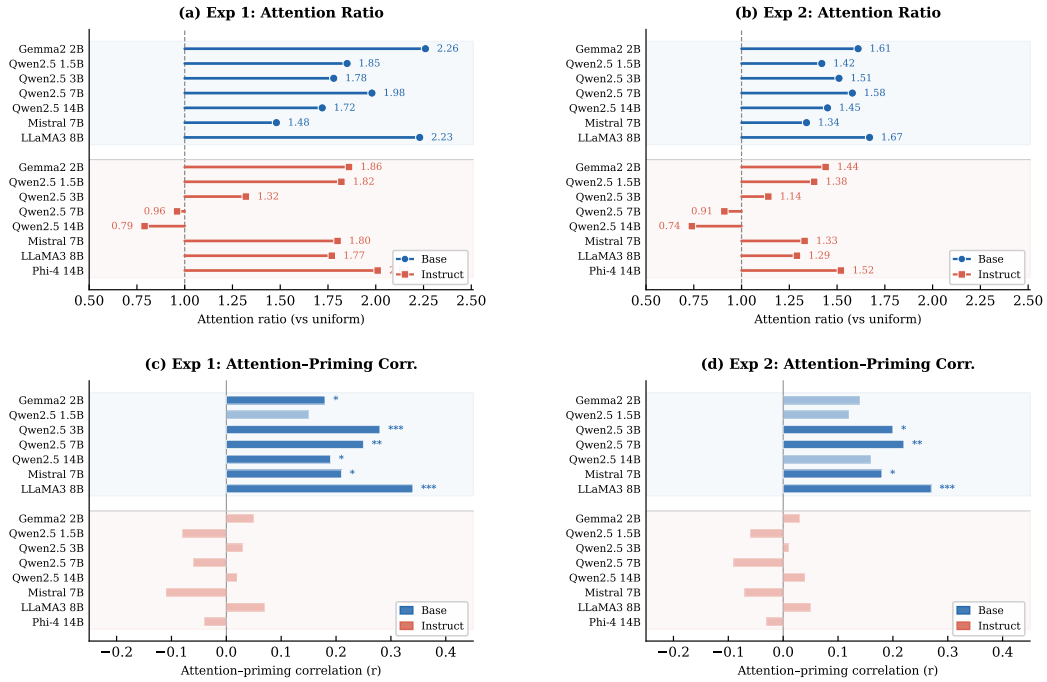


Figure 4: Attention analysis. **(a–b)** Attention ratio to prior occurrences vs. uniform baseline (dashed line). Base models show elevated attention; instruct models decline with scale. **(c–d)** Attention–priming correlation. Base models show significant positive correlations; instruct models show weak correlations.

and non-significant correlations ($|r| < 0.12$), suggesting that attention-based retrieval is decoupled from the downstream processing that determines the behavioral outcome.

Scaling trend. Within the Qwen2.5 instruct family, attention ratio declined monotonically with scale across both tasks, with the largest models allocating *less* attention to prior occurrences than expected by chance. This suggests that at sufficient scale, the post-training pipeline may begin to suppress attention-based retrieval itself.

5.3 Evidence 3: Targeted Attention-Head Ablation

We ranked attention heads by how strongly their attention to prior target occurrences predicted item-level priming, then ablated the top 10 during inference. Base-model priming dropped sharply (LLaMA3: $+6.89 \rightarrow +2.62$; Qwen2.5: $+5.52 \rightarrow +1.84$), whereas random-head controls largely preserved it ($+6.12$ and $+5.21$). The same targeted intervention barely changed instruct models (LLaMA3: $+2.05 \rightarrow +1.97$; Qwen2.5: $+2.31 \rightarrow +2.14$). Thus prior-occurrence attention is causally involved in base-model facilitation, whereas instruct behavior is comparatively insensitive to this pathway (Appendix Table 4).

6 Discussion and Conclusion

The results map onto the *automatic* vs. *controlled processing* framework (Shiffrin & Schneider, 1977). Base models exhibit **automatic processing**: repeated words trigger lag-invariant facilitation that partly survives context removal and label remapping. Instruct models exhibit **controlled processing**: context-dependent re-evaluation that decays with lag and collapses without expected context. Correlational attention results and causal head ablations converge: prior-occurrence retrieval directly supports base-model priming, but is filtered or gated before shaping instruct outputs. The source of that gating remains open. It may reflect conflict resolution or broader post-training-induced repetition avoidance, not necessarily

RLHF alone; isolating SFT from preference optimization requires checkpoints along a controlled post-training trajectory. Human processing is lag-sensitive like instruct models but uniformly facilitative like base models (Bowers, 2000; Tenpenny, 1995). Practically, RAG, few-shot prompting, and dialogue all reuse contextual information; our results predict that the effects of repeated chunks, examples, or entities depend on model type and recency. Repetition priming therefore reveals a consequential change associated with post-training while also identifying a hybrid human profile that neither model type captures.

Ethics Statement

The human experiments in this study were approved by the Institutional Review Board (IRB) of Yale University. All participants provided written informed consent prior to participation and were compensated at a rate of $15/hour. The experiments involved only standard psycholinguistic tasks (word categorization and sentence completion) with no deceptive procedures, sensitive content, or foreseeable risks beyond those of everyday computer use. All human data were collected and stored in accordance with institutional data protection guidelines and are anonymized in all analyses.

LLM Usage Disclosure

An LLM-based writing assistant was used to support language editing and camera-ready organization. All scientific claims, analyses, and final text were reviewed and verified by the authors.

References

Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. What learning algorithm is in-context learning? investigations with linear models. In *International Conference on Learning Representations*, 2023.

Jeffrey S Bowers. In defense of abstractionist theories of repetition priming and word identification. *Psychonomic Bulletin & Review*, 7(1):83–99, 2000.

Oliver Brady, Paul Nulty, Lili Zhang, Tomás E Ward, and David P McGovern. Dual-process theory and decision-making in large language models. *Nature Reviews Psychology*, 4: 777–792, 2025.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pp. 1877–1901, 2020.

Florin Cuconasu, Giovanni Trappolini, Nicola Tonellotto, and Fabrizio Silvestri. A tale of trust and accuracy: Base vs. instruct LLMs in RAG systems. *arXiv preprint arXiv:2406.14972*, 2024.

Damai Dai, Yutao Sun, Li Dong, Yaru Hao, Shuming Ma, Zhifang Sui, and Furu Wei. Why can GPT learn in-context? language models secretly perform gradient descent as meta-optimizers. In *Findings of the Association for Computational Linguistics: ACL 2023*, pp. 4005–4019, 2023.

Xufeng Duan, Bei Xiao, Xuemei Tang, and Zhenguang G Cai. HLB: Benchmarking LLMs' humanlikeness in language use. *arXiv preprint arXiv:2409.15890*, 2024.

Allyson Ettinger, Ahmed Elgohary, and Philip Resnik. Probing for semantic evidence of composition by means of simple classification tasks. In *Proceedings of the 1st Workshop on Evaluating Vector-Space Representations for NLP*, pp. 134–139, 2016.

Kenneth I Forster and Chris Davis. Repetition priming and frequency attenuation in lexical access. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 10(4):680–698, 1984.

Kalanit Grill-Spector, Richard Henson, and Alex Martin. Repetition and the brain: neural models of stimulus-specific effects. *Trends in Cognitive Sciences*, 10(1):14–23, 2006.

Richard N Henson. Neuroimaging studies of priming. *Progress in Neurobiology*, 70(1):53–81, 2003.

Richard N Henson, Anthony Rylands, Elinor Ross, Patrik Vuilleumier, and Michael D Rugg. The effect of repetition lag on electrophysiological and haemodynamic correlates of visual object priming. *NeuroImage*, 21(4):1674–1689, 2004.

Zhengbao Jiang, Zhiqing Sun, Weijia Shi, Pedro Rodriguez, Chunting Zhou, Graham Neubig, Xi Victoria Lin, Wen-tau Yih, and Srinivasan Iyer. Instruction-tuned language models are better knowledge learners. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, 2011.

Zhong-Zhi Li, Duzhen Zhang, Ming-Liang Zhang, Jiaxin Zhang, Zengyan Liu, Yuxuan Yao, Haotian Xu, Junhao Zheng, Pei-Jie Wang, Xiuyi Chen, et al. From system 1 to system 2: A survey of reasoning large language models. *arXiv preprint arXiv:2502.17419*, 2025.

Jiachang Liu, Dinghan Shen, Yizhe Zhang, Bill Dolan, Lawrence Carin, and Weizhu Chen. What makes good in-context examples for GPT-3? In *Proceedings of Deep Learning Inside Out (DeeLIO 2022): The 3rd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures*, pp. 100–114, 2022.

Matéo Mahaut and Francesca Franzon. Repetitions are not all alike: Distinct mechanisms sustain repetition in language models. *arXiv preprint arXiv:2504.01100*, 2025. URL https://arxiv.org/abs/2504.01100.

Elinor McKone. Short-term implicit memory for words and nonwords. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 21(5):1108–1126, 1995.

Ken McRae, George S Cree, Mark S Seidenberg, and Chris McNorgan. Semantic feature production norms for a large set of living and nonliving things. *Behavior Research Methods*, 37(4):547–559, 2005.

Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. Rethinking the role of demonstrations: What makes in-context learning work? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 11048–11064, 2022.

Kanishka Misra, Allyson Ettinger, and Julia Taylor Rayz. Exploring BERT's sensitivity to lexical cues using tests from semantic priming. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 4625–4635, 2020.

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.

Jonathan E Peelle, Ryland L Miller, Chad S Rogers, Brent Spehar, Mitchell S Sommers, and Kristin J Van Engen. Completion norms for 3085 English sentence contexts. *Behavior Research Methods*, 52:1795–1799, 2020.

Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. Language models as knowledge bases? In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 2463–2473, 2019.

Giuseppe Sartori and Graziella Orrù. Language models and psychological sciences. *Frontiers in Psychology*, 14:1279317, 2023.

Don L Scarborough, Charles Cortese, and Hollis S Scarborough. Frequency and repetition effects in lexical memory. *Journal of Experimental Psychology: Human Perception and Performance*, 3(1):1–17, 1977.

Richard M Shiffrin and Walter Schneider. Controlled and automatic human information processing: II. Perceptual learning, automatic attending, and a general theory. *Psychological Review*, 84(2):127–190, 1977.

Arabella Sinclair, Jaap Jumelet, Willem Zuidema, and Raquel Fernandez. Structural persistence in language models: Priming as a window into abstract language representations. *Transactions of the Association for Computational Linguistics*, 10:1031–1050, 2022.

Wilson L Taylor. Cloze procedure: A new tool for measuring readability. *Journalism Quarterly*, 30(4):415–433, 1953.

Patricia L Tenpenny. Abstractionist versus episodic theories of repetition priming and word identification. *Psychonomic Bulletin & Review*, 2(3):339–363, 1995.

Aditya R. Vaidya, Javier S. Turek, and Alexander G. Huth. Humans and language models diverge when predicting repeating text. In *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*, pp. 58–69. Association for Computational Linguistics, 2023. URL https://aclanthology.org/2023.conll-1.5/.

Xuansheng Wu, Wenlin Yao, Jianshu Chen, Xiaoman Pan, Xiaoyang Wang, Ninghao Liu, and Dong Yu. From language modeling to instruction following: Understanding the behavior shift in LLMs after instruction tuning. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 2341–2369, 2024.

Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. Calibrate before use: Improving few-shot performance of language models. In *International Conference on Machine Learning*, pp. 12697–12706. PMLR, 2021.

Alireza S Ziabari, Nona Ghazizadeh, Zhivar Sourati, Farzan Karimi-Malekabadi, Payam Piray, and Morteza Dehghani. Reasoning on a spectrum: Aligning LLMs to system 1 and system 2 thinking. *arXiv preprint arXiv:2502.12470*, 2025.

A Extended Related Work

Dual-Process Frameworks for LLMs. The distinction between fast, heuristic processing (System 1) and slow, deliberate reasoning (System 2) (Kahneman, 2011) has been increasingly applied to LLMs. Brady et al. (2025) argue that LLMs exhibit both System-1-like biases and System-2-like reasoning depending on prompting strategy. Li et al. (2025) survey the rapid development of reasoning LLMs (e.g., OpenAI o1, DeepSeek R1), framing progress as a shift from System 1 to System 2 capabilities. Ziabari et al. (2025) propose methods to align LLMs along this spectrum. However, these approaches primarily manipulate the *prompting* or *training* regime to elicit different reasoning modes. Our work differs in showing that the dual-process distinction emerges *spontaneously* from instruction tuning, observable through a simple repetition priming paradigm without any explicit prompting for fast or slow reasoning.

LLM–Human Cognitive Comparisons. A growing body of work evaluates LLMs as models of human cognition. Sartori & Orrù (2023) provide a broad assessment of language models as tools for psychological science, noting both striking correspondences and systematic divergences. Duan et al. (2024) benchmark LLMs on humanlikeness across multiple linguistic dimensions. In psycholinguistics specifically, LLMs replicate many human behavioral patterns including semantic priming and garden-path effects (Misra et al., 2020; Ettinger et al., 2016), but also show non-human patterns such as insensitivity to word frequency effects that are robust in human data. Our matched human experiments contribute to this literature by identifying a specific phenomenon, repetition priming, where human behavior falls between two distinct LLM processing modes, rather than aligning with either.

Effects of Instruction Tuning on Internal Representations. Beyond behavioral changes, instruction tuning reshapes model internals. Wu et al. (2024) demonstrate that instruction tuning modifies self-attention to capture instruction-verb relationships and rotates feed-forward knowledge toward task-relevant directions. Mechanistic interpretability studies show that fine-tuning surfaces pre-existing capabilities rather than constructing entirely new circuits (Olsson et al., 2022). Cross-model steering experiments suggest shared representational subspaces between base and instruct variants. Our attention analysis (Section 5.2) complements these findings by showing that while both model types share an attention-based retrieval mechanism for prior occurrences, instruction tuning decouples this retrieval from downstream behavioral effects.

Repetition Suppression in Neuroscience. In cognitive neuroscience, repeated stimulus presentation typically produces *repetition suppression* (reduced neural responses to familiar stimuli) alongside behavioral facilitation (Henson, 2003; Henson et al., 2004). This dissociation between neural suppression and behavioral facilitation parallels our finding that instruct models can show reduced attention to prior occurrences (especially at scale) while still exhibiting some behavioral facilitation. The neuroscience literature also emphasizes that repetition effects are modulated by attention, expectation, and recognition (Grill-Spector et al., 2006), variables that map onto the controlled processing characteristics we observe in instruct models.

Distinct Repetition Mechanisms. Vaidya et al. (2023) show that human and LM predictions diverge for repeating passages and relate the LM pattern to middle-layer attention heads and recency. Mahaut & Franzon (2025) further distinguish natural repetition from ICL-induced repetition: the two patterns differ in confidence, specialized heads, and sensitivity to perturbations. Our task differs from both by measuring controlled repetition priming rather than repeated generation, but the conceptual convergence is important: repetition is not a unitary mechanism, and superficially similar effects can depend on different attention pathways and contextual demands.

B Experimental Design Details

Lag Manipulation. In both experiments, we manipulated the number of intervening trials between successive presentations of a target word. Table 2 summarizes the key design parameters. For Experiment 1, lag values of 1, 3, 7, and 15 were tested; for Experiment 2, lag values of 1, 3, and 7 were tested, plus an isolated baseline session. Each lag condition was run as a separate session with fresh context.

Trial Structure. In both experiments, each target word was tested in an independent context block: context was reset before each word's first exposure, ensuring priming effects reflect only within-word repetition. In Experiment 1, each trial presented a single word for classification, with 45 filler words providing intervening trials. In Experiment 2, each trial presented a sentence with a blank to be completed.

Parameter	Exp 1	Exp 2
Target words	182	21
Fillers	45	50
Presentations per word	4 (E1–E4)	4 (E0–E3)
Lag conditions	1, 3, 7, 15	1, 3, 7
Cloze range	—	0.05–0.15
Context isolation	per word	per word
Sessions per model	728	84

Table 2: Experimental design parameters.

C Stimuli

Experiment 1. 182 target words were selected from the McRae feature norms (91 living, 91 nonliving). Table 3 shows representative examples.

Category	Examples
Living (91)	tiger, elephant, artist, baby, child, robin, salmon, …
Nonliving (91)	bus, car, bicycle, bowl, jacket, hammer, violin, …

Table 3: Target words for Experiment 1 (182 words total, selected from McRae feature norms).

Experiment 2. The 21 target words were: *church, fear, fight, friend, horse, laugh, lure, mud, names, nose, rash, rat, ring, sail, salt, song, sour, speed, spider, steal, wash*. Each was paired with 4 low-cloze sentence frames (cloze 0.05–0.15). Example for *fight*: (E0) "The man was professionally trained in how to _____." (cloze = 0.13), (E1) "Afraid of getting injured, Zach backed out of the _____." (cloze = 0.14), (E2) "The boys were told not to _____." (cloze = 0.08), (E3) "They ended up having a huge _____." (cloze = 0.11).

D Prompt Formats

For Experiment 1, the prompt-sensitivity analysis used the following three semantically equivalent templates:

Template 1 (original)

```
Task: Classify each word as 'living' or 'nonliving'.
Word: tiger
Answer: living
Word: bus
Answer:
```

Template 2

```
Decide whether each word denotes a living thing or a nonliving thing.
Word: tiger
Category: living
Word: bus
Category:
```

Template 3

```
For each item, choose exactly one label: living or nonliving.
Item: tiger
Label: living
Item: bus
Label:
```

Instruct models received each version wrapped in the appropriate chat template; response labels, trial history, item order, and scored tokens were otherwise unchanged. For Experi-

ment 2: `Complete the sentence with one word: The group of friends would often ___. Answer: laugh`

Prompt-sensitivity control. We evaluated three semantically equivalent Experiment 1 templates that varied the instruction wording and answer cue while preserving the response options. Across templates, the LLaMA3-Base lag coefficient ranged from -0.04 to $+0.01$ (all n.s.), whereas LLaMA3-Instruct ranged from -0.32 to -0.27 (all $p < .001$). The Prompt$\times$ModelType$\times$Lag interaction was not significant ($F = 1.12$, $p = .33$), indicating that wording differences did not explain the dissociation.

Ablation Condition. Responses replaced with "..." placeholders.

Label Remapping. Abstract labels (A/B) instead of semantic labels. Phase 1: 20 words $\times$ 3 exposures with M1. Mapping change + 8 warmup trials. Test: 20 TARGET vs. 20 CONTROL words under M2.

E Model Specifications

All experiments used HuggingFace Transformers (v4.36.0) with bfloat16 precision, eager attention, greedy decoding, and 100-trial context limit. Hardware: NVIDIA A100 80GB GPUs.

F Statistical Analysis Details

Priming Effect. $\text{Priming}_{E_i} = \text{Margin}_{E_i} - \text{Margin}_{E_1}$. We report priming separately for each exposure (E2$-$E1, E3$-$E1, E4$-$E1) as well as the mean across E2–E4, averaged across all target words and lag conditions.

Omnibus models and multiplicity. Within each experiment, an omnibus mixed-effects model included ModelType, Exposure, Lag, and their interactions as fixed effects, with random intercepts for Model and Word. The ModelType$\times$Lag interaction was significant in Experiment 1 ($\beta = -0.26$, $SE = 0.04$, $t = -6.50$, $p < .001$) and Experiment 2 ($\beta = -0.13$, $SE = 0.03$, $t = -4.52$, $p < .001$). False-discovery-rate correction preserved negative lag effects for $5/8$ instruct models in Experiment 1 and $6/8$ in Experiment 2, versus $0/7$ base models in either experiment.

Effect sizes. Standardized between-type contrasts were large: Experiment 1 mean priming $d = 2.18$, Experiment 1 final priming $d = 1.82$, Experiment 2 mean priming $d = 2.90$, and the missing-context contrast $d = 2.93$. In label remapping, LLaMA3-Base retained a $+1.00$ TARGET-over-CONTROL margin after remapping ($p < .01$, $d = 0.87$).

Attention Analysis. Attention to prior occurrences: (1) identify positions of prior presentations, (2) sum attention weights from final token, (3) average across heads per layer, (4) compute ratio to uniform-distribution expectation.

Targeted head ablation. For LLaMA3-8B and Qwen2.5-7B base/instruct pairs, we ranked individual heads by the association between attention to earlier target occurrences and item-level priming, selected the top 10, and ablated their outputs during inference. Random sets of 10 heads served as controls. Table 4 reports final priming under each condition.

Model	Full	Top-10	Random-10
LLaMA3 Base	+6.89	+2.62	+6.12
LLaMA3 Instruct	+2.05	+1.97	+2.01
Qwen2.5 Base	+5.52	+1.84	+5.21
Qwen2.5 Instruct	+2.31	+2.14	+2.25

Table 4: Targeted attention-head ablation in Experiment 1. Ablating heads most associated with prior-occurrence retrieval reduces base-model priming, whereas instruct models and random-head controls change little.

G Human–Model Comparison Details

The human and model dependent measures have different units and should not be compared in absolute magnitude. Human RT/accuracy/production serve as a cognitive reference for the qualitative shape of the effect.

Pattern	Base LMs	Instruct LMs	Humans
Facilitation	Strong positive	Weaker/variable	Positive RT/accuracy
Lag sensitivity	None	Predominantly negative	Significant in RT
Long-lag interference	Not observed	Some models	Not observed

Table 5: Profile-level comparison. Human behavioral measures and model log-probabilities are used to compare direction and lag dependence, not metric-level magnitudes.

Experiment	Measure	Initial/baseline	Final
Semantic categorization	RT	648 ms	539 ms
	Accuracy	92.3%	97.1%
Cloze completion	Target production	11.2%	41.7%
	Correct-production RT	4.82 s	3.41 s

Table 6: Human performance summaries. Experiment 1 compares E1 with E4. Cloze production compares the no-study baseline with three exposures; RT compares E1 with E3 among correctly produced targets.

H Additional Results

Model	Size	E2−E1	E3−E1	E4−E1	Mean
Base models					
LLaMA3	8B	+3.94	+6.82	+6.89	+5.88
Qwen2.5-7B	7B	+3.12	+5.50	+5.52	+4.71
Qwen2.5-3B	3B	+2.95	+5.42	+5.59	+4.65
Mistral	7B	+2.99	+5.42	+5.34	+4.58
Qwen2.5-14B	14B	+2.91	+5.21	+5.32	+4.48
Qwen2.5-1.5B	1.5B	+2.68	+5.10	+5.18	+4.32
Gemma2	2B	+2.64	+4.99	+5.04	+4.22
Instruct models					
Phi-4	14B	+2.21	+3.41	+3.80	+3.14
Qwen2.5-1.5B	1.5B	+1.59	+3.24	+3.78	+2.87
Gemma2	2B	+1.34	+2.72	+3.21	+2.42
Qwen2.5-3B	3B	+1.05	+2.58	+3.31	+2.31
Qwen2.5-7B	7B	+0.73	+1.82	+2.31	+1.62
LLaMA3	8B	+0.80	+1.64	+2.05	+1.50
Mistral	7B	−0.10	−0.92	−1.63	−0.88
Qwen2.5-14B	14B	−1.41	−2.98	−4.14	−2.84

Table 7: Priming by exposure in Experiment 1 (change in log-probability margin from E1 baseline, averaged across lag conditions).

Model	Size	Exposure		Lag	
		β	t	β	t
Base models					
Qwen2.5-3B	3B	+1.62	11.24***	−0.01	−0.34
LLaMA3	8B	+1.94	11.98***	−0.03	−1.07
Qwen2.5-1.5B	1.5B	+1.52	8.41***	+0.02	0.87
Qwen2.5-7B	7B	+1.58	11.75***	−0.03	−1.08
Mistral	7B	+1.56	10.57***	−0.01	−0.28
Gemma2	2B	+1.48	8.79***	−0.04	−1.28
Qwen2.5-14B	14B	+1.55	9.82***	−0.01	−0.42
Instruct models					
Qwen2.5-7B	7B	+0.62	1.38	−0.18	−3.24**
Qwen2.5-3B	3B	+0.88	1.72	−0.11	−1.92
Gemma2	2B	+1.18	3.26**	−0.14	−1.48
Qwen2.5-1.5B	1.5B	+1.08	2.82**	−0.09	−2.10*
Phi-4	14B	+1.01	2.40*	−0.13	−3.08**
LLaMA3	8B	+0.57	1.25	−0.30	−6.74***
Mistral	7B	−0.14	−0.39	−0.34	−11.36***
Qwen2.5-14B	14B	−0.97	−1.59	−0.21	−3.84***

Table 8: Mixed-effects model results (Experiment 1). Exposure β: change per additional exposure. Lag β: change per unit lag increase. *$p < .05$, **$p < .01$, ***$p < .001$.

Model	Size	E1−E0	E2−E0	E3−E0	Mean
Base models					
LLaMA3	8B	+2.41	+3.87	+4.12	+3.47
Gemma2	2B	+2.18	+3.52	+3.71	+3.14
Mistral	7B	+1.82	+3.01	+3.24	+2.69
Qwen2.5-14B	14B	+1.54	+2.68	+2.95	+2.39
Qwen2.5-7B	7B	+1.45	+2.51	+2.78	+2.25
Qwen2.5-3B	3B	+1.21	+2.14	+2.42	+1.92
Qwen2.5-1.5B	1.5B	+0.98	+1.82	+2.11	+1.64
Instruct models					
Gemma2	2B	+0.92	+1.54	+1.85	+1.44
Qwen2.5-1.5B	1.5B	+0.88	+1.47	+1.78	+1.38
Phi-4	14B	+0.81	+1.38	+1.62	+1.27
Mistral	7B	+0.72	+1.18	+1.41	+1.10
Qwen2.5-3B	3B	+0.65	+1.08	+1.32	+1.02
Qwen2.5-7B	7B	+0.58	+0.95	+1.14	+0.89
LLaMA3	8B	+0.51	+0.84	+1.02	+0.79
Qwen2.5-14B	14B	+0.35	+0.52	+0.61	+0.49

Table 9: Priming by exposure in Experiment 2 (change in target log-probability from E0 baseline, averaged across lag conditions).

Model	Size	Exposure		Lag	
		β	t	β	t
Base models					
LLaMA3	8B	+1.37	9.84***	−0.02	−0.71
Gemma2	2B	+1.24	8.92***	−0.01	−0.38
Mistral	7B	+1.08	7.64***	−0.03	−1.15
Qwen2.5-14B	14B	+0.98	7.12***	−0.02	−0.82
Qwen2.5-7B	7B	+0.93	6.85***	−0.02	−0.76
Qwen2.5-3B	3B	+0.81	5.94***	−0.01	−0.42
Qwen2.5-1.5B	1.5B	+0.70	4.87***	−0.01	−0.35
Instruct models					
Gemma2	2B	+0.62	3.41***	−0.11	−2.42*
Qwen2.5-1.5B	1.5B	+0.59	3.18**	−0.10	−2.18*
Phi-4	14B	+0.54	2.87**	−0.12	−2.64**
Mistral	7B	+0.47	2.54*	−0.15	−3.28**
Qwen2.5-3B	3B	+0.44	2.38*	−0.13	−2.87**
Qwen2.5-7B	7B	+0.38	2.05*	−0.16	−3.41***
LLaMA3	8B	+0.34	1.82	−0.18	−3.95***
Qwen2.5-14B	14B	+0.20	1.08	−0.21	−4.12***

Table 10: Mixed-effects model results (Experiment 2). Same model specification as Experiment 1 (Equation 3). *$p < .05$, **$p < .01$, ***$p < .001$.

Model	Lag 1	Lag 3	Lag 7	Lag 15
Base models				
Gemma2-Base	+0.38	+0.42	+0.44	+0.40
LLaMA3-Base	+1.52	+1.41	+1.34	+1.24
Mistral-Base	+0.95	+0.88	+0.81	+0.74
Qwen2.5-1.5B-Base	+0.45	+0.42	+0.39	+0.37
Qwen2.5-3B-Base	+0.62	+0.58	+0.54	+0.50
Qwen2.5-7B-Base	+0.78	+0.72	+0.65	+0.58
Qwen2.5-14B-Base	+0.71	+0.66	+0.60	+0.55
Instruct models				
Gemma2-Inst	−3.39	−2.69	−1.48	−1.15
LLaMA3-Inst	−1.86	−1.17	−2.82	−1.69
Mistral-Inst	−3.23	−2.42	−1.11	−0.94
Phi4-Inst	−1.24	−1.01	−1.43	−1.36
Qwen2.5-1.5B-Inst	−2.75	−2.65	−1.41	−1.99
Qwen2.5-3B-Inst	−6.40	−4.40	−2.80	−2.39
Qwen2.5-7B-Inst	−7.33	−3.29	−3.66	−5.07
Qwen2.5-14B-Inst	+2.49	+0.35	−2.97	−2.31

Table 11: No-response ablation by lag (Exp 1). Priming effect when model responses are replaced with placeholder tokens. Base models show weak positive effects; instruct models show strong negative effects. Qwen2.5-14B-Inst showed positive priming at lag 1 (+2.49) reversing to negative at longer lags.

Phase	**Mean Margin**	**SD**	**Accuracy**	n
Baseline (E0)	0.98	0.72	90%	20
Exposure 1	4.21	2.15	95%	20
Exposure 2	6.87	1.89	100%	20
Exposure 3	7.89	1.52	100%	20
Test phase (after label remapping)				
Condition	**Mean Margin**	**SD**	**Accuracy**	n
TARGET (seen in P1)	5.02	1.34	100%	20
CONTROL (new words)	4.02	1.21	100%	20
Difference: TARGET − CONTROL = +1.00, $p < .01$, $d = 0.87$				

Table 12: Label remapping results (LLaMA3-Base). Top: Phase 1 learning curve showing rapid margin increase. Bottom: test phase under remapped labels (M2). TARGET words show higher margins than CONTROL, indicating representational facilitation.